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Case Study 17: Online Movie Ticket Booking

1. Introduction

The Online Movie Ticket Booking System is a simple console-based program developed using C++. The system allows users to view available cinema seats, book seats, and generate a ticket confirmation.

The program simulates a basic movie ticket reservation system where seats are managed using a two-dimensional array. It helps users understand how programming concepts like arrays, loops, and conditional statements can be applied to real-world problems.

2. Objectives

The objectives of this case study are:

- To design a simple movie ticket booking system.
- To display available and booked seats.
- To allow users to select and book a seat.
- To generate a ticket confirmation message after booking.
- To understand array-based seat management.

3. Technologies Used

C++ Programming Language

The program is developed using C++ and executed in a console environment.

Header File

`#include <iostream>` is used for input and output operations.

4. Concepts Used

The following programming concepts are used in the system:

- **Two-Dimensional Arrays** – Used to represent rows and columns of cinema seats.
- **Loops (for loop)** – Used to display the seat layout.
- **Conditional Statements (if-else)** – Used to check seat availability and validate input.
- **Switch Case** – Used to create a menu-driven program.
- **User Input and Output** – Using cin and cout to interact with the user.
- **Menu Driven Programming** – Allows users to select different operations.

5. System Features

The Online Movie Ticket Booking System includes the following features:

Show Available Seats

Displays the seat layout where:

- **0** represents an available seat
- **1** represents a booked seat

Book Ticket

Allows the user to select a **row and seat number** to book a seat.

Print Ticket

After booking a seat, the system prints a ticket confirmation containing:

- Movie name
- Seat number
- Booking status

6. Working of the Program

1. The system displays a menu with three options:
    - Show Available Seats
    - Book Ticket
    - Exit
2. When the user selects Show Available Seats, the program displays the seat layout.
3. When the user selects Book Ticket, the user enters the row and seat number.
4. The system checks:
    - If the seat selection is valid.
    - If the seat is already booked.
5. If the seat is available, the system marks it as booked and prints the ticket confirmation.
6. The program continues running until the user selects Exit.

7. Output

```cpp
        cout << "===== ONLINE MOVIE TICKET BOOKING =====" << endl;
        cout << "1. Show Available Seats" << endl;
        cout << "2. Book Ticket" << endl;
        cout << "3. Exit" << endl;
        cout << "=====================================" << endl;
        cout << "Enter your choice: ";
        cin >> choice;

        switch(choice) {

            case 1:
                cout << "Seat Layout (0 = Available, 1 = Booked):" << endl;
                for(int i = 0; i < rows; i++) {
                    cout << "Row " << char(65 + i) << " : ";
                    for(int j = 0; j < cols; j++) {
                        cout << seats[i][j] << " ";
                    }
                    cout << endl;
                }
                break;

            case 2:
                cout << "Enter Row (A-D): ";
                cin >> row;
                cout << "Enter Seat Number (1-5): ";
                cin >> col;

                if(row < 1 || row > rows || col < 1 || col > cols) {
                    cout << "Invalid seat selection!" << endl;
                }
                else if(seats[row-1][col-1] == 1) {
                    cout << "Seat already booked!" << endl;
```

```
atharva@Atharvas-MacBook-Air-5 Main_projectc++ % g++ main.cpp -o m
atharva@Atharvas-MacBook-Air-5 Main_projectc++ % ./m
===== ONLINE MOVIE TICKET BOOKING =====
1. Show Available Seats
2. Book Ticket
3. Exit
=====================================
Enter your choice: 1
Seat Layout (0 = Available, 1 = Booked):
Row A : 0 0 0 0 0
Row B : 0 0 0 0 0
Row C : 0 0 0 0 0
Row D : 0 0 0 0 0
===== ONLINE MOVIE TICKET BOOKING =====
1. Show Available Seats
2. Book Ticket
3. Exit
=====================================
Enter your choice:
```

```cpp
        cout << "2. Book Ticket" << endl;
        cout << "3. Exit" << endl;
        cout << "=====================================" << endl;
        cout << "Enter your choice: ";
        cin >> choice;

        switch(choice) {

            case 1:
                cout << "Seat Layout (0 = Available, 1 = Booked):" << endl;
                for(int i = 0; i < rows; i++) {
                    cout << "Row " << char(65 + i) << " : ";
                    for(int j = 0; j < cols; j++) {
                        cout << seats[i][j] << " ";
                    }
                    cout << endl;
                }
                break;

            case 2:
                cout << "Enter Row (A-D): ";
                cin >> row;
                cout << "Enter Seat Number (1-5): ";
                cin >> col;

                if(row < 1 || row > rows || col < 1 || col > cols) {
                    cout << "Invalid seat selection!" << endl;
                }
                else if(seats[row-1][col-1] == 1) {
                    cout << "Seat already booked!" << endl;
```

```
atharva@Atharvas-MacBook-Air-5 Main_projectc++ % ./m
Seat Layout (0 = Available, 1 = Booked):
Row A : 0 0 0 0 0
Row B : 0 0 0 0 0
Row C : 0 0 0 0 0
Row D : 0 0 0 0 0
===== ONLINE MOVIE TICKET BOOKING =====
1. Show Available Seats
2. Book Ticket
3. Exit
=====================================
Enter your choice: 2
Enter Row (A-D): 1
Enter Seat Number (1-5): 3
===== TICKET CONFIRMED =====
Movie: Dhurandhar
Seat: Row A Seat 3
Status: Confirmed
===== ONLINE MOVIE TICKET BOOKING =====
1. Show Available Seats
2. Book Ticket
3. Exit
=====================================
Enter your choice:
```

8. Learning Outcomes

From this case study, the following concepts were learned:

- Implementation of array-based seat management
- Creating menu-driven programs
- Using loops and conditions for seat validation
- Understanding basic ticket booking logic

9. Conclusion

The Online Movie Ticket Booking System demonstrates how C++ programming can be used to create a simple reservation system. By using arrays to manage seats and implementing menu-driven logic, the program provides a basic simulation of real-world movie ticket booking systems.

This project helps strengthen fundamental programming concepts and problem-solving skills.